

AP CSP | Lesson 17: Lists & Abstraction

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: A

A copies the earlier value 4. B treats assignment as a live link. C invents addition. D invents a reset.

Q2. Answer: B

B is the second element. A uses the first position; C the third; D confuses index with value.

Q3. Answer: C

C replaces position 2. A inserts; B updates position 1; D appends. Update preserves length.

Q4. Answer: D

D accesses the sole remaining value 5. The new length is 1; indices 0, 2, 5 are invalid.

Q5. Answer: A and C

A counts zero elements. C adds one element at the end. B mistakes an index for a searched value. D contradicts ordered lists, which may contain duplicates.

Q6. Answer: B

B identifies a concrete abstraction benefit. Lists do not guarantee speed, data correctness or valid indices, so A,C,D are unsupported.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) 5; (b) 12; (c) 4; (d) 1, 2, 3, 4.

1 mark: First value 5.

1 mark: Third value 12.

1 mark: Length 4.

1 mark: Exactly indices 1 through 4.

Q8. 4 marks

Answer: (a) [10, 20, 30, 40]. (b) [10, 15, 20, 30, 40]. (c) [10, 15, 20, 40]. (d) [5, 15, 20, 40].

1 mark: Correct list after APPEND.

1 mark: Correct list after INSERT, shifting old indices 2 onward right.

1 mark: Correct list after removing current index 4.

1 mark: Correct list after replacement at index 1.

Q9. 4 marks

Answer: (a) [4, 6, 12]. (b) [4, 6, 8, 12]. (c) [4, 8, 12, 6]. (d) [4, 12].

1 mark: Correct update list.

1 mark: Correct insertion list.

1 mark: Correct append list.

1 mark: Correct removal list.

Q10. 4 marks

Answer: sessions ← [15, 30]
APPEND(sessions, 45)
sessions[1] ← 20
DISPLAY(sessions[LENGTH(sessions)])

1 mark: Creates the specified list.

1 mark: APPEND adds 45 at the end.

1 mark: Updates index 1 to 20.

1 mark: Displays the element at current LENGTH, not a hard-coded last index.

Q11. 4 marks

Answer: (a) [30]. (b) 1. (c) 30. (d) An invalid-index error terminates execution.

1 mark: List[30].

1 mark: Length 1.

1 mark: First value 30.

1 mark: Index 0 causes an error and program termination.

Q12. 4 marks

Answer: (a) sessions. (b) [10, 20, 30]. (c) APPEND(sessions,newValue) adds an entry to the collection. (d) For example, entries must still be checked for valid minute values; a list does not guarantee correctness.

1 mark: Meaningful name tied to sessions/minutes.

1 mark: Correct ordered list literal[10, 20, 30].

1 mark: Explains APPEND or equivalent collection operation avoids another separate variable.

1 mark: Identifies a genuine remaining requirement, such as valid data or valid indices, without claiming guaranteed speed.